

Market Cluster Dynamics — S&P 500 Prototype

(2026-09-28, rev. 2)

The problem with rev. 1

The stationary-drift model (rev. 1, 2026-09-27) reproduced calm-market distributions (RMSE 0.045) but could not generate the observed spikes: March 2021 saw 55% of stocks pile into the top momentum bin (>60% twelve-month return); late 2008 saw 98% pile into the bottom bin (<0%). A monthly Markov transition matrix mean-reverts too fast to build or hold such concentrations. Herding (bilinear flux toward above-average-momentum bins) was a weak amplifier at best — it can only redistribute mass the drift already moves.

New reaction kernels (the mechanism first)

Three kernels were added. Each has a physical mechanism, not just a fitted parameter.

1. Market advection (the spike generator)

Mechanism. When the market's trailing twelve-month return shifts by M in a month, every stock's momentum shifts by $\sim M$ in return space — a rigid translation of the whole distribution along the momentum axis. This happens mechanically at window rollover: the month entering the 12-month window replaces the month dropping out. The driver is

$$M(t) = r(t) - r(t-12)$$

the *synchronized change in twelve-month momentum*, measured cross-sectionally (equal-weighted over all index stocks, M_{cs}). March 2021: $M_{cs} = +0.256$ (the -12% March-2020 crash month exited the window as the +15% March-2021 month entered). October 2008: $M_{cs} = -0.256$.

Implementation. Donor-cell advection within each volatility lane: a shift M moves fraction $f = \kappa |M|/w_{eff}$ of each bin to the next, $\kappa = 1.0$ (the mechanical value, not fitted), $w_{eff} = 0.15$ assumed (typical momentum-bin

width). Boundaries reflect: the extreme bins (<0%, >60%) are unbounded, so mass that reaches them accumulates — this is what generates spikes. M_{cs} is measured from Yahoo Finance daily closes.*

Timing note. The transition $t_0 \rightarrow t_1$ uses the *destination-month* driver $M(t_1)$, because the window rollover that defines $M(t_1)$ is what moves stocks between bins during that month. This makes the hindcast *conditional* on the realized market move — an exogenous-forcing run, not a lookahead-free forecast. A genuine forecast needs a forecast (or scenario) for $M(t_1)$.

2. Autocatalytic herding (the amplifier)

Mechanism. Winners attract flows; flows push prices; prices make winners. The herding rate should grow with the winner concentration itself — autocatalysis, not a constant rate.

Implementation. $h_{eff} = h_0 + h_1(w - l)$, where w, l are the top/bottom momentum shares. $h_0 = 0$, $h_1 = 1.0$ calibrated* (one-step hindcast error on pre-episode data: 2000-2019 for the 2020-21 case, 2000-2007 for the 2008-09 case). The signed form is deliberate: in a crash ($l \gg w$) it drives mass toward losers (panic selling); in a melt-up ($w \gg l$) toward winners (FOMO). It is symmetric by construction — the data decide the sign.

3. Sticky drift (the persistence kernel)

Mechanism. A concentrated distribution decays slower: when 98% of stocks are losers, the "average" transition matrix (estimated over all regimes) overstates the escape rate, because it mixes crash months with normal months.

Implementation. $T_{eff} = (1-\lambda)T + \lambda I$, with $\lambda = \min(\lambda_1 \max(w, l), 0.9)$, $\lambda_1 = 1.0$ calibrated* (same pre-episode protocol). When concentration is low, ordinary drift; when the distribution piles into an extreme bin, the drift freezes toward identity and the spike persists.

Master equation

$$n(t+1) = T_{eff}' n(t) + \text{advect}(n(t), M_{cs}) + h_{eff} * H(n(t)) + s - d * n(t)$$

Drift (measured T , sticky), advection (measured M_{cs}), herding (calibrated), entry/exit (measured).

Calibration (pre-episode only — no peeking)

episode	T window	h1, lam1 fit on
2020-21	2000-2019 (measured)	one-step error, 2015-2019
2008-09	2000-2007 (measured)	one-step error, 2005-2007

kappa = 1.0 is mechanical (a 15pp market move shifts momentum 15pp = one bin width); w_eff = 0.15 assumed. No parameter was fit on the spike months themselves.

Results — the spikes are reproduced

2020-21 (COVID crash + recovery)

- Peak: **actual 0.546 @ Mar-2021 → predicted 0.579 @ Apr-2021** (amplitude error +6%, one month late).
- Full-distribution RMSE: **0.0915** (rev. 1 best: 0.0821 — slightly worse on RMSE, but rev. 1 never produced any spike at all: its peak was 0.09 vs the actual 0.55).
- Decay after peak: actual 0.41/0.33/0.36 @ +1/+2/+3 mo; predicted 0.26/0.17/0.16. **Too fast** — see limitations.

2008-09 (financial crisis)

- Peak: **actual 0.978 @ Mar-2009 → predicted 0.983 @ Nov-2008** (amplitude error +0.5%; the actual is a plateau from late 2008 through Mar-2009 — the model captures its onset and height).
- Full-distribution RMSE: **0.1182** (rev. 1 best: 0.1110).
- Decay after peak: actual 0.97/0.94/0.88; predicted 0.98/0.61/0.72. +1 mo is exact; then too fast — see limitations.

What changed vs rev. 1

	rev. 1 (regime T, h=0.2)	rev. 2 (new kernels)
2021 peak	0.09 (no spike)	0.58 (actual 0.55)
2008 peak	0.30 (no spike)	0.98 (actual 0.98)

The advection kernel is doing the work: the March-2021 spike is the +25.6pp window-delta mechanically translating the distribution into the unbounded top bin; the 2008 spike is the -25.6pp October delta translating it into the bottom bin. Herding amplifies (+0.16 on the 2008 peak at $h_1=1$ vs $h_1=0$) and sticky drift holds the plateaus.

Limitations (stated plainly)

1. **Decay too fast.** After each peak the model drains the extreme bin $\sim 2x$ faster than observed. Mechanism: the donor-cell moves a uniform fraction f of the bin, but at a real peak the bin's mass sits deep in the tail (Mar-2021: median top-bin momentum was 103%, only 11% within 8pp of the 60% boundary — measured). A 15-bin distribution cannot resolve within-bin depth without memory of how the mass got there. Fixing this needs either sub-binning the extremes or an age-structured extreme bin.
2. **Conditional, not lookahead-free.** The hindcast uses the realized $M(t_1)$. It answers "given the market moved M , where does the distribution go?" — the forcing is exogenous. A live forecast needs a market-move forecast alongside.
3. **2021 peak one month late.** The model puts the top-bin maximum in Apr-2021 vs the actual Mar-2021 — the mechanical translation lags the true (faster-than-monthly) repricing by one step.
4. **2008 plateau too short.** The model holds 0.98 for ~ 2 months; the actual plateau lasted ~ 5 . Same within-bin-depth cause as (1).

Data provenance

- Prices: Yahoo Finance daily closes, S&P 500 constituents, 2000-2021 (**measured**; survivorship bias noted — current constituents backfilled).
- Index additions/removals: entry/exit rates **measured** per episode.
- M_{cs} : cross-sectional window-delta **measured** from the same prices.
- T : monthly transition matrices **measured** per calibration window.
- $\kappa=1.0$: mechanical identity (a pp is a pp). $w_{eff}=0.15$ **assumed**.
- $h_1=1.0$, $\lambda_1=1.0$: **calibrated** on pre-episode one-step errors.
- Spike-threshold/tail experiments (concentration-dependent extreme-bin outflow): tested, did not improve both episodes jointly — dropped.

Next steps

1. Sub-bin or age-structure the extreme bins to fix the decay rate.
2. Forecast $M(t+1)$ (VIX term structure? options-implied?) to make the hindcast a genuine forecast.
3. Test on a third episode (2000-02 dot-com, 2022 rate shock).
4. Push to GitHub (branch Stock-Market-Dynamics) once approved.

References

- Yahoo Finance (finance.yahoo.com) — daily OHLCV, S&P 500 constituents; used for momentum/volatility bins, transition matrices, and the M_{cs} driver.
- CBOE VIX, FRED series VIXCLS — regime labels in rev. 1 experiments.
- Project code: `market_dynamics/src/model.py` (master equation), `src/hindcast_final.py` (frozen-config runs).